

A METHOD FOR OBTAINING GENERATING FUNCTIONS OF CONDITIONAL DISTRIBUTIONS AND SOME APPLICATIONS

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1. Introduction

There are many cases in the field of multivariate distribution theory where it is easier to use the moment generating function (m.g.f.) or the probability generating function (p.g.f.) as starting point instead of the probability function. Further, it is well-known that if for k random variables the m.g.f. $M(a_1, a_2, \dots, a_k)$ or the p.g.f. $G(t_1, t_2, \dots, t_k)$ is given then the m.g.f. or p.g.f. of the marginal distribution of any subset of the variables follows from $M(a_1, a_2, \dots, a_k)$ or $G(t_1, t_2, \dots, t_k)$ by putting the $a_i = 0$ or $t_i = 1$ for all values of " i " corresponding to variables not included in the subset.

In this paper a straightforward method is given for obtaining the m.g.f. or p.g.f. of conditional distributions. Although the method is based only on a restatement of the well-known theorem of compound probabilities, the authors are of opinion that the value of the method is not generally realised. In multivariate statistical theory it often works when other methods fail. (see examples 2.2 and 2.5.)

2. The Generating Functions of Conditional Distributions

(i) Random Variables

Given k random variables X_i , $i = 1, 2, \dots, k$ defined for $a_i \leq x_i \leq b_i$ and with probability density function $f(x_1, x_2, \dots, x_k)$ and m.g.f.

$$M(a_1, a_2, \dots, a_k) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_k}^{b_k} \exp\left(\sum_{i=1}^k a_i x_i\right) f(x_1, x_2, \dots, x_k) \prod_{i=1}^k dx_i,$$

and with marginal distribution of $X_{r+1}, X_{r+2}, \dots, X_k$:

$$f_{r+1,k}(x_{r+1}, x_{r+2}, \dots, x_k) = \int_{a_1}^{b_1} \int_{a_2}^{b_2} \dots \int_{a_r}^{b_r} f(x_1, x_2, \dots, x_k) \prod_{i=1}^r dx_i,$$

then it follows immediately from

$$f(x_1, x_2, \dots, x_k) = f_{r+1,k}(x_{r+1}, x_{r+2}, \dots, x_k) f(x_1, x_2, \dots, x_r | x_{r+1}, x_{r+2}, \dots, x_k)$$

that

$$(2.1) \quad M(a_1, a_2, \dots, a_k) = \int_{a_{r+1}}^{b_{r+1}} \int_{a_{r+2}}^{b_{r+2}} \dots \int_{a_k}^{b_k} \exp\left(\sum_{i=r+1}^k a_i x_i\right) f_{r+1,k}(x_{r+1}, x_{r+2}, \dots, x_k) \cdot \Phi(a_1, a_2, \dots, a_r, x_{r+1}, x_{r+2}, \dots, x_k) \prod_{i=r+1}^k dx_i$$

where $\Phi(a_1, a_2, \dots, a_r, x_{r+1}, x_{r+2}, \dots, x_k)$ is the m.g.f. of the conditional distribution of $X_1, X_2, \dots, X_r$ given $X_{r+1} = x_{r+1}, X_{r+2} = x_{r+2}, \dots, X_k = x_k$.

The equation (2.1) also holds for discrete distributions if the integration signs are changed to summation signs.

In vector notation, equation (2.1) can be written as

$$(2.2) \quad M(a^{(1)}, d^{(2)}) = \int \exp(a^{(2)'} x^{(2)}) f(x^{(2)}) \Phi(a^{(1)}, x^{(2)}) dx^{(2)}$$

where $\Phi(a^{(1)}, x^{(2)})$ is the m.g.f. of the conditional distribution of the vector $X^{(1)}$ given that the vector $X^{(2)} = x^{(2)}$ and where $f(x^{(2)})$ is the marginal distribution of $X^{(2)}$. A similar result follows for more than two vectors.

Similarly, the p.g.f. can also be written in such a form that, when the p.g.f. together with the marginal distribution is known, the p.g.f. of the conditional distribution follows immediately and vice versa, i.e. for discrete distributions:

$$(2.3) \quad G(t_1, t_2, \dots, t_k) = \sum_{x_{r+1}} \sum_{x_{r+2}} \dots \sum_{x_k} f_{r+1, k}(x_{r+1}, x_{r+2}, \dots, x_k) \cdot t_{r+1}^{x_{r+1}} t_{r+2}^{x_{r+2}} \dots t_k^{x_k} \cdot \psi(t_1, t_2, \dots, t_r, x_{r+1}, x_{r+2}, \dots, x_k)$$

where $\psi(t_1, t_2, \dots, t_r, x_{r+1}, x_{r+2}, \dots, x_k)$ is the p.g.f. of the conditional distribution of $X_1, X_2, \dots, X_r$ given $X_{r+1} = x_{r+1}, X_{r+2} = x_{r+2}, \dots, X_k = x_k$.

Example 2.1

If the k mutually exclusive events $E_1, E_2, \dots, E_k$ may occur in each trial with probability $p_1, p_2, \dots, p_k$, respectively, and if

$$p_0 = 1 - \sum_{i=1}^k p_i$$

is the probability that not one of the E_i 's ($i = 1, 2, \dots, k$) occur, then the probability for obtaining x_i successes of E_i ($i = 1, 2, \dots, k$) after n trials are made, is given by the coefficient of $t_1^{x_1} t_2^{x_2} \dots t_k^{x_k}$ in the expansion of

$$(2.4) \quad (p_0 + p_1 t_1 + p_2 t_2 + \dots + p_k t_k)^n$$

and is the well-known multinomial probability function.

Clearly, (2.4) is equal to

$$(2.5) \quad \sum_{x_{r+1}} \sum_{x_{r+2}} \dots \sum_{x_k} \left\{ \frac{n!}{x_{r+1}! x_{r+2}! \dots x_k! (n - \sum_{i=r+1}^k x_i)!} p_{r+1}^{x_{r+1}} p_{r+2}^{x_{r+2}} \dots p_k^{x_k} \cdot (1 - \sum_{i=r+1}^k p_i)^{(n - \sum_{i=r+1}^k x_i)} \right\} t_{r+1}^{x_{r+1}} t_{r+2}^{x_{r+2}} \dots t_k^{x_k} \cdot [(p_0 + p_1 t_1 + \dots + p_r t_r) / (1 - \sum_{i=r+1}^k p_i)]^{(n - \sum_{i=r+1}^k x_i)}$$

where the factor in square brackets is the marginal distribution of $X_{r+1}, X_{r+2}, \dots, X_k$. Therefore (2.5) is in the required form (2.2) from which it follows that

$$[(p_0 + p_1 t_1 + \dots + p_r t_r) / (1 - \sum_{i=r+1}^k p_i)]^{(n - \sum_{i=r+1}^k x_i)}$$

is the p.g.f. of $X_1, X_2, \dots, X_r$ given $X_{r+1} = x_{r+1}, X_{r+2} = x_{r+2}, \dots, X_k = x_k$.

Example 2.2

The above result follows, of course, fairly easily by first obtaining the conditional distribution by ordinary methods. The value of the method may be better illustrated by considering a more complex case where the joint probability function is difficult to work with, but the marginal distributions are simple. As a further example, consider the case of three events that can occur separately or jointly in each trial. Let the p.g.f. now be $G(t_1, t_2, t_3) = (p_{000} + p_{100}t_1 + p_{010}t_2 + p_{001}t_3 + p_{110}t_1t_2 + p_{011}t_1t_3 + p_{101}t_2t_3 + p_{111}t_1t_2t_3)^n$ where p_{ijl} is the probability for E_1, E_2, E_3 to occur i, j and l times respectively.

The p.g.f. of the marginal distribution of X_1 is thus given by

$$(q + pt_1)^n \text{ where } p = p_{100} + p_{110} + p_{101} + p_{111} \text{ and } q = 1 - p.$$

Clearly, by taking terms containing t_1 together,

$$G(t_1, t_2, t_3) = \sum_{x_1=0}^n \frac{n!}{(n-x_1)! x_1!} p^{x_1} q^{n-x_1} t_1^{x_1} \cdot \left[\{ (p_{100} + p_{110}t_2 + p_{101}t_3 + p_{111}t_2t_3)^{x_1} \cdot (p_{000} + p_{010}t_2 + p_{001}t_3 + p_{011}t_2t_3)^{n-x_1} \} / \{ p^{x_1} q^{n-x_1} \} \right]$$

The function inside the square brackets is thus the p.g.f. of X_2 and X_3 given $X_1 = x_1$

Example 2.3

It is sometimes convenient to define the multivariate normal distribution as that distribution for which all cumulants of the third and higher order are zero. The m.g.f. of the multivariate normal distribution is now considered from this angle by dividing the vector a into two vectors $a^{(1)}: p_1 \times 1$ and $a^{(2)}: p_2 \times 1$. By using the ordinary notation, the mg.f. is given by (Anderson (1958)).

$$\begin{aligned} M(a^{(1)}, a^{(2)}) &= \exp \left\{ \frac{1}{2} (a^{(1)})' \Sigma_{11} a^{(1)} + 2a^{(1)'} \Sigma_{12} a^{(2)} + a^{(2)'} \Sigma_{22} a^{(2)} \right\} \\ &= \exp \left\{ \frac{1}{2} (a^{(2)} + \Sigma_{22}^{-1} \Sigma_{21} a^{(1)})' \Sigma_{22} (a^{(2)} + \Sigma_{22}^{-1} \Sigma_{21} a^{(1)}) \right\} \cdot \\ &\quad \cdot \exp \left\{ \frac{1}{2} (a^{(1)'} \Sigma_{11} a^{(1)} - a^{(1)'} \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21} a^{(1)}) \right\}. \end{aligned}$$

Using

$$\exp\left\{\frac{1}{2}(\beta' \Sigma_{22} \beta)\right\} = \frac{1}{|\Sigma_{22}|^{\frac{1}{2}} (2\pi)^{\frac{1}{2}} p_2} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2}x^{(2)'} \Sigma_{22}^{-1} x^{(2)}\right\}$$

$$\exp\{x^{(2)'} \beta\} dx^{(2)}, \text{ where } \beta = a^{(2)} + \Sigma_{22}^{-1} \Sigma_{21} a^{(1)},$$

it follows that,

$$M(a^{(1)}, a^{(2)}) = \int_{-\infty}^{\infty} \left[\frac{1}{|\Sigma_{22}|^{\frac{1}{2}} (2\pi)^{\frac{1}{2}} p_2} \exp\left\{-\frac{1}{2}x^{(2)'} \Sigma_{22}^{-1} x^{(2)}\right\} \right] \exp\{x^{(2)'} a^{(2)}\} \cdot \\ (2.6) \quad \cdot [\exp\{x^{(2)'} \Sigma_{22}^{-1} \Sigma_{21} a^{(1)} + \frac{1}{2} a^{(1)'} (\Sigma_{11} - \Sigma_{12} \Sigma_{22}^{-1} \Sigma_{21}) a^{(1)}\}] dx^{(2)},$$

where the first factor of the integrand is the marginal distribution of $X^{(2)}$, where the random vector X is split into two vectors $X^{(1)} : p_1 \times 1$ and $X^{(2)} : p_2 \times 1$. Thus

$$\exp\{x^{(2)'} \Sigma_{22}^{-1} \Sigma_{21} a^{(1)} + \frac{1}{2} a^{(1)'} \Sigma_{11.2} a^{(1)}\}$$

is the m.g.f. of the conditional distribution of $X^{(1)}$ given $X^{(2)} = x^{(2)}$.

Example 2.4

It was shown by Steyn (1955) that the multivariate hypergeometric function can be used as p.g.f. for a wide class of discrete distributions. For this class of distributions the p.g.f. is

$$(2.7) \quad C F(a; b_1, b_2, \dots, b_k; c; t_1, t_2, \dots, t_k)$$

$$= C \sum_{x_i=0}^{\infty} \frac{(a + \sum_{i=1}^k x_i - 1) \left(\sum_{i=1}^k x_i \right)!}{k!} \frac{\prod_{i=1}^k (b_i + x_i - 1) (x_i)!}{x_i!} \frac{t_i^{x_i}}{x_i!} \\ i=1, \dots, k \quad (c + \sum_{i=1}^k x_i - 1) \left(\sum_{i=1}^k x_i \right)!$$

where $r^{(s)} = r(r-1) \dots (r-s+1)$ and

$$C = \frac{1}{F(a; \sum_{i=1}^k b_i; c; 1)}$$

Using the fact that the p.g.f. of the marginal distribution of $X_{r+1}, X_{r+2}, \dots, X_k$ is

obtained from (2.7) by putting $t_1 = t_2 = \dots = t_r = 1$, it follows that (2.7) is equal to

$$\sum_{x_{r+1}} \sum_{x_{r+2}} \dots \sum_{x_k} f_{r+1, k}(x_{r+1}, x_{r+2}, \dots, x_k) \frac{t_{r+1}^{x_{r+1}}}{x_{r+1}!} \frac{t_{r+2}^{x_{r+2}}}{x_{r+2}!} \dots \frac{t_k^{x_k}}{x_k!}.$$

$$(2.8) \quad \left[\frac{F(a + \sum_{i=r+1}^k x_i; b_1, b_2, \dots, b_r; c + \sum_{i=r+1}^k x_i; t_1, t_2, \dots, t_r)}{F(a + \sum_{i=r+1}^k x_i; \sum_{i=1}^r b_i; c + \sum_{i=r+1}^k x_i; 1)} \right]$$

which is the required form (2.3) and shows that the function inside the square brackets is the p.g.f. of the conditional distribution of $X_1, X_2, \dots, X_r$ given $X_{r+1} = x_{r+1}, X_{r+2} = x_{r+2}, \dots, X_k = x_k$.

The multivariate hypergeometric function given in (2.7) is usually referred to as a F_1 type. There are also other types of hypergeometric functions that can be used as generating functions. (Stein (1957)).

Similarly, there are cases where multivariate hypergeometric functions appear as the m.g.f. of a continuous multivariate distribution. The fact that multivariate hypergeometric functions are easily split as in (2.8) makes it very convenient in all such cases to find conditional distributions. The Dirichlet distribution (Wilks (1962)) is an example of such a case.

(ii) Random Matrices

In the case of random matrices, $X_{(i)}$, $i = 1, 2, \dots, k$, equation (2.1) can be written as

$$(2.9) \quad M(a_{(1)}, a_{(2)}, \dots, a_{(k)}) = \int \dots \int \exp\left(\sum_{i=r+1}^k a_{(i)}' x_{(i)}\right) \cdot f_{r+1, k}(x_{(r+1)}, x_{(r+2)}, \dots, x_{(k)}) \cdot \Phi(a_{(1)}, a_{(2)}, \dots, a_{(r)}; x_{(r+1)}, x_{(r+2)}, \dots, x_{(k)}) \prod_{i=r+1}^k dx_{(i)}$$

where

$$\Phi(a_{(1)}, a_{(2)}, \dots, a_{(r)}; x_{(r+1)}, x_{(r+2)}, \dots, x_{(k)})$$

is the m.g.f. of the conditional distribution of $X_{(1)}, X_{(2)}, \dots, X_{(r)}$ given

$$X_{(r+1)} = x_{(r+1)}, X_{(r+2)} = x_{(r+2)}, \dots, X_{(k)} = x_{(k)} \text{ and } f_{r+1, k}(x_{(r+1)}, x_{(r+2)}, \dots, x_{(k)})$$

is the marginal distribution of $X_{(r+1)}, X_{(r+2)}, \dots, X_{(k)}$.

Example 2.5

The m.g.f. for the non-central Wishart matrix A when sampling from multivariate normal populations with $\Sigma = I$, is given by (Herz (1955)) as

$$M(a) = E(\exp\{\text{tr } A a\}) = \exp\{\text{tr } T a (I - 2a)^{-1}\} \cdot |I - 2a|^{-\frac{1}{2}n},$$

where T is the noncentrality parameter.

Consider now the case where the matrices $a : 2p \times 2p$, $A : 2p \times 2p$ and $T : 2p \times 2p$ are as follows:

$$a = \begin{pmatrix} a_{(11)} & a_{(12)} \\ a_{(21)} & a_{(22)} \end{pmatrix}, \quad A = \begin{pmatrix} A_{(11)} & A_{(12)} \\ A_{(21)} & A_{(22)} \end{pmatrix}, \quad T = \begin{pmatrix} T_{(11)} & T_{(12)} \\ T_{(21)} & O \end{pmatrix}$$

so that $A_{(22)} : p \times p$ has a central Wishart distribution. After some algebraic manipulations using that $\text{tr } T(a(1-2a))^{-1} = \text{tr} \{ -\frac{1}{2}T + \frac{1}{2}T(I-2a)^{-1} \}$ it follows that

$$\begin{aligned}
 & M(a_{(1)}, a_{(12)}, a_{(21)}, a_{(22)}) \\
 &= \exp \left\{ -\frac{1}{2} \text{tr } T_{(11)} \right\} \cdot \exp \left\{ \frac{1}{2} \text{tr } T_{(11)} (I-2a_{(1)})^{-1} \right\} \cdot |I-2a_{(1)}|^{-\frac{1}{2}n} \\
 &\cdot |I-2a_{(22)} - 4a_{(21)}(I-2a_{(11)})^{-1}a_{(12)}|^{-\frac{1}{2}n} \\
 &\cdot \exp \left\{ \frac{1}{2} \text{tr} \left[\left\{ 2T_{(21)}(I-2a_{(11)})^{-1}a_{(12)} + 2a_{(21)}(I-2a_{(11)})^{-1}T_{(12)} \right. \right. \right. \\
 &\quad \left. \left. \left. + 4a_{(21)}(I-2a_{(11)})^{-1}T_{(11)}(I-2a_{(11)})^{-1}a_{(12)} \right\} \left\{ I-2a_{(22)} - 4a_{(21)}(I-2a_{(11)})^{-1}a_{(12)} \right\}^{-1} \right] \right\} \\
 &= \exp \left\{ \frac{1}{2} \text{tr } T_{(11)} [(I-2a_{(11)})^{-1}I] \right\} \cdot |I-2a_{(11)}|^{-\frac{1}{2}n} \\
 &\cdot |I-2a_{(22)} - 4a_{(21)}(I-2a_{(11)})^{-1}a_{(12)}|^{-\frac{1}{2}n} \\
 &\cdot \sum_{k=0}^{\infty} \sum_K C_K \left[\left\{ T_{(21)}(I-2a_{(11)})^{-1}a_{(12)} + a_{(21)}(I-2a_{(11)})^{-1}T_{(12)} \right. \right. \\
 &\quad \left. \left. + 2a_{(21)}(I-2a_{(11)})^{-1}T_{(11)}(I-2a_{(11)})^{-1}a_{(12)} \right\} \right. \\
 &\cdot \left. \left. \left\{ I-2a_{(22)} - 4a_{(21)}(I-2a_{(11)})^{-1}a_{(12)} \right\}^{-1} \right] (k!)^{-1}
 \end{aligned}$$

where $\exp \{ \text{tr } S \} = \sum_{k=0}^{\infty} \sum_K C_K(S) (k!)^{-1}$ and $C_K(S)$ a zonal polynomial.

By using the integral

$$\int_{S \geq 0} \exp \left\{ -\frac{1}{2} \text{tr } RS \right\} |S|^{\frac{1}{2}(n-p-1)} C_K \left(\frac{1}{2} S P \right) dS = 2^{\frac{1}{2}np} \Gamma_p \left(\frac{1}{2}n, K \right) |R|^{-\frac{1}{2}n} C_K(P R^{-1})$$

where $S: p \times p$ matrix and $\Gamma_p \left(\frac{1}{2}n, K \right) = \pi^{\frac{1}{2}p} \Gamma_p(p-1) \frac{\pi^p}{i!} \Gamma \left(\frac{1}{2}n + K, \frac{1}{2}(i-1) \right)$,

as derived by Constantine (1963), it follows for $|R| = |I-2a_{(22)} - 4a_{(21)}(I-2a_{(11)})^{-1}a_{(12)}|$

that,

$$\begin{aligned}
 (2.10) \quad & M(a_{(1)}, a_{(12)}, a_{(21)}, a_{(22)}) \\
 &= \int_{A_{(22)} > 0} \left[2^{-\frac{1}{2}np} (\Gamma_p \left(\frac{1}{2}n \right))^{-1} \exp \left\{ -\frac{1}{2} \text{tr } A_{(22)} \right\} |A_{(22)}|^{-\frac{1}{2}(n-p-1)} \right] \exp \{ \text{tr } A_{(22)} a_{(22)} \} \\
 &\cdot \left[\exp \left\{ \frac{1}{2} \text{tr } T_{(11)} \left\{ (I-2a_{(11)})^{-1}I \right\} \right\} \cdot |I-2a_{(11)}|^{-\frac{1}{2}n} \right. \\
 &\cdot \left. \exp \left\{ 2 \text{tr } A_{(22)} a_{(21)} (I-2a_{(11)})^{-1} a_{(12)} \right\} \right]
 \end{aligned}$$

$$\cdot {}_0F_1 \left[\frac{1}{2}n; \frac{1}{2}A_{(22)} \left\{ T_{(21)}(I-2a_{(11)})^{-1}a_{(12)} + a_{(21)}(I-2a_{(11)})^{-1}T_{(12)} \right. \right. \\ \left. \left. + 2a_{(21)}(I-2a_{(11)})^{-1}T_{(11)}(I-2a_{(11)})^{-1}a_{(12)} \right\} \right] dA_{(22)}$$

$$\text{where } {}_0F_1(\frac{1}{2}n;S) = \sum_{k=0}^{\infty} \sum_K \frac{\Gamma_p(\frac{1}{2}n)}{\Gamma_p(\frac{1}{2}n,K)} C_K(S) (k!)^{-1}$$

The function inside the first square bracket of (2.10) is the marginal distribution of $A_{(22)}$ viz. $W(I,n)$. Thus (2.10) is in the required form (2.9), from which it follows that the factor in the last square brackets is the m.g.f. of the conditional distribution of $A_{(11)}, A_{(12)}$ and $A_{(21)}$ given $A_{(22)}$.

3. Applications

A form of application follows by remembering that K. Pearson (1903) considered, for a normal multivariate population, the influence of normal selection with respect to one or more variables on the variances, correlations and regression coefficients of the remaining variables. The investigation by Pearson implies that the marginal distribution for some of the normal variables undergoes a change in parameters but remains normal, while the conditional distribution of the remaining variables for given values of the selected variables remains unchanged. Pearson's results for normal selection may easily be written down by changing the parameters of

$$\frac{1}{|\Sigma_{22}|^{\frac{1}{2}} (2\pi)^{\frac{1}{2}p_2}} \exp\left\{-\frac{1}{2}x^{(2)'} \Sigma_{22}^{-1} x^{(2)}\right\}$$

in (2.6) and by studying the new joint m.g.f. (Aitken, 1934).

Having noticed the above analogy for normal selection, it is now clear that the same ideas and equation (2.1) may be used in the case of non-normal continuous distributions and in the case of discrete multivariate distributions. There are practical problems where the influence of a change in the parameters of the marginal distribution is of importance.

Example 3.1

Each member of a population of n voters may vote for any one of k candidates $A_1, A_2, \dots, A_k$ or may destroy his ballot paper if he does not favour any of the candidates. All the candidates are interested in the distribution of the number of votes in their favour as they lose their deposit if they do not draw a certain minimum percentage of the votes. Now consider the following situation. The candidates $A_j, j=1, 2, \dots, r$ know the conditional distribution of the votes in their favour if the number of votes for $A_j, j=r+1, \dots, k$ is known. Let $p_j, j=1, \dots, k$ be the known probability at the start of the election campaign that a voter will vote for A_j . During the campaign the probability p_j changes to $p'_j, j=r+1, \dots, k$. What is the influence of this change on the joint distribution and expectations of $X_j, j=1, 2, \dots, r$?

Clearly, the joint p.g.f. of $X_j, j=1, \dots, k$ is given by (2.4) and (2.5). The influence of a change in the marginal distribution of $X_j, j=r+1, \dots, k$ such that the conditional distribution of $X_j, j=1, 2, \dots, r$ remains unchanged is given by changing p_j to $p'_j, j=r+1, \dots, k$ in (2.5), thereby obtaining:

$$\begin{aligned}
& \sum_{x_r+1} \dots \sum_{x_k} \frac{n!}{x_{r+1}! x_{r+2}! \dots x_k! (n - \sum_{j=r+1}^k x_j)!} p_{r+1}^{x_{r+1}} p_{r+2}^{x_{r+2}} \dots p_k^{x_k} \\
& \cdot (1 - \sum_{j=r+1}^k p_j)^{(n - \sum_{j=r+1}^k x_j)} t_{r+1}^{x_{r+1}} t_{r+2}^{x_{r+2}} \dots t_k^{x_k} \\
& \cdot (p_0 + p_1 t_1 + \dots + p_r t_r) / (1 - \sum_{j=r+1}^k p_j)^{(n - \sum_{j=r+1}^k x_j)} \\
& = (R p_0 + R p_1 t_1 + R p_2 t_2 + \dots + R t_r + p'_{r+1} t_{r+1} + \dots + p'_k t_k)^n
\end{aligned}$$

where $R = (1 - \sum_{j=r+1}^k p_j) / (1 - \sum_{j=r+1}^k p_j)$ so that $E(X_i) = n R p_i, i=1, 2, \dots, r$.

This was meant to be straightforward example with a result that could have been written down more or less by direct reasoning and was meant to make the procedure clear.

It was already clear from section 2 of this paper that a number of the well-known multivariate distributions – continuous and discrete – can easily be written in the form (2.1) and (2.3). In these cases it is also possible to study the influence of a change in marginal distributions.

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